Unsupervised Learning

A Project Report

Submitted to

greatlearning
Learning for Life

By

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Context

AllLife Bank wants to focus on its credit card customer base in the next financial year. They have been advised by their marketing research team, that the penetration in the market can be improved. Based on this input, the Marketing team proposes to run personalized campaigns to target new customers as well as upsell to existing customers. Another insight from the market research was that the customers perceive the support services of the back poorly. Based on this, the Operations team wants to upgrade the service delivery model, to ensure that customer queries are resolved faster. The Head of Marketing and Head of Delivery both decide to reach out to the Data Science team for help.

Objective

To identify different segments in the existing customers, based on their spending patterns as well as past interaction with the bank, using clustering algorithms, and provide recommendations to the bank on how to better market to and service these customers.

Data Description

The data provided is of various customers of a bank and their financial attributes like credit limit, the total number of credit cards the customer has, and different channels through which customers have contacted the bank for any queries (including visiting the bank, online, and through a call center).

Column Name	Description
Sl_No	Primary key of the records
Customer Key	Customer identification number
Average Credit Limit	Average credit limit of each customer for all credit cards
Total credit cards	Total number of credit cards possessed by the customer
Total visits bank	Total number of visits that the customer made (yearly) personally to the bank
Total visits online	Total number of visits or online logins made by the customer (yearly)
Total calls made	Total number of calls made by the customer to the bank or its customer service department (yearly)

Table 1: Data Description

Data Overview

1. Import the required packages.
2. Loaded the dataset using **read_csv()**.
3. Used the **head()** to get the first five rows of the data.

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	1	87073	100000	2	1	1	0
1	2	38414	50000	3	0	10	9
2	3	17341	50000	7	1	3	4
3	4	40496	30000	5	1	1	4
4	5	47437	100000	6	0	12	3

Table 2. First Five Rows Of Dataset

4. Used the **tail()** to get the last five rows of the data.

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
655	656	51108	99000	10	1	10	0
656	657	60732	84000	10	1	13	2
657	658	53834	145000	8	1	9	1
658	659	80655	172000	10	1	15	0
659	660	80150	167000	9	0	12	2

Table 3: Last Five Rows Of Dataset

5. Checked the shape of the dataset using the **shape method.**
    a. The number of rows is 660.
    b. The number of columns is 7.
6. Copied the dataframe to another variable.
7. Checked the datatype of all the columns in the dataset using info().
    a. There are 7 rows and all the columns belong to integer type.
    b. There are no missing values.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 660 entries, 0 to 659
Data columns (total 7 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Sl_No                660 non-null    int64
 1   Customer Key         660 non-null    int64
 2   Avg_Credit_Limit     660 non-null    int64
 3   Total_Credit_Cards   660 non-null    int64
 4   Total_visits_bank    660 non-null    int64
 5   Total_visits_online  660 non-null    int64
 6   Total_calls_made     660 non-null    int64
dtypes: int64(7)
memory usage: 36.2 KB
```

Fig 1: Datatype Of Each Column

8. There are no duplicate rows present in the dataframe checked using duplicated().
9. There are no missing values present in the dataframe check using isnull().
10. Removed Unwanted columns from the dataframe(Sl_No , Customer Key) using **drop()**.
11. Checked the statistical description for the dataset

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
count	660.000000	660.000000	660.000000	660.000000	660.000000
mean	34574.242424	4.706061	2.403030	2.606061	3.583333
std	37625.487804	2.167835	1.631813	2.935724	2.865317
min	3000.000000	1.000000	0.000000	0.000000	0.000000
25%	10000.000000	3.000000	1.000000	1.000000	1.000000
50%	18000.000000	5.000000	2.000000	2.000000	3.000000
75%	48000.000000	6.000000	4.000000	4.000000	5.000000
max	200000.000000	10.000000	5.000000	15.000000	10.000000

Table 4: Statistical Description For The Dataset

    a. Most people have a credit limit around ₹18,000, but a few have very high limits up to ₹2,00,000.
    b. Usually, customers keep 3 to 6 credit cards.
    c. Bank visits are low mostly between 1 to 4 times, some people use it up to 5 times.
    d. Online visits are mostly between 1 to 4 times, some people use it up to 15 times.
    e. Calls to the bank are mostly 1 to 5, but some make up to 10 calls.
12. Checked the number of unique values present in each column.
    a. In the Avg_Credit_Limit column there are 110 unique values.
    b. In the Total_Credit_Cards column there are 10 unique values.
    c. In the Total_Visits_bank column there are 6 unique values.
    d. In the Total_Visits_online column there are 16 unique values.
    e. In the Total_calls_made column there are 11 unique values.

Exploratory Data Analysis

Univariate Analysis

1.1. Observations on Avg_Credit_Limit

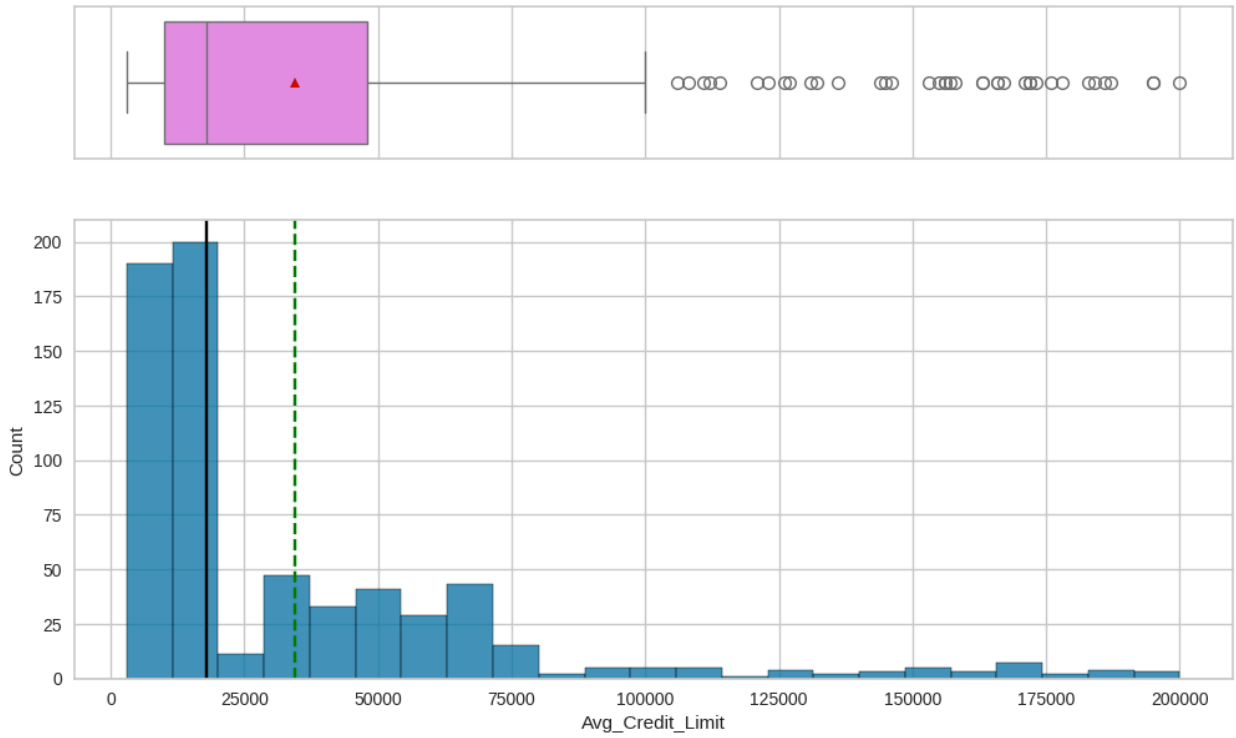

Fig 2: Observation on Avg_Credit_Limit

Observations:

- Most people have a credit limit below ₹50,000.
- A few customers have very high limits up to ₹2,00,000.
- The chart is pulled to the right, showing higher values for some people.
- There are many outliers with very high credit limits.

1.2. Observations on Total_Credit_Cards

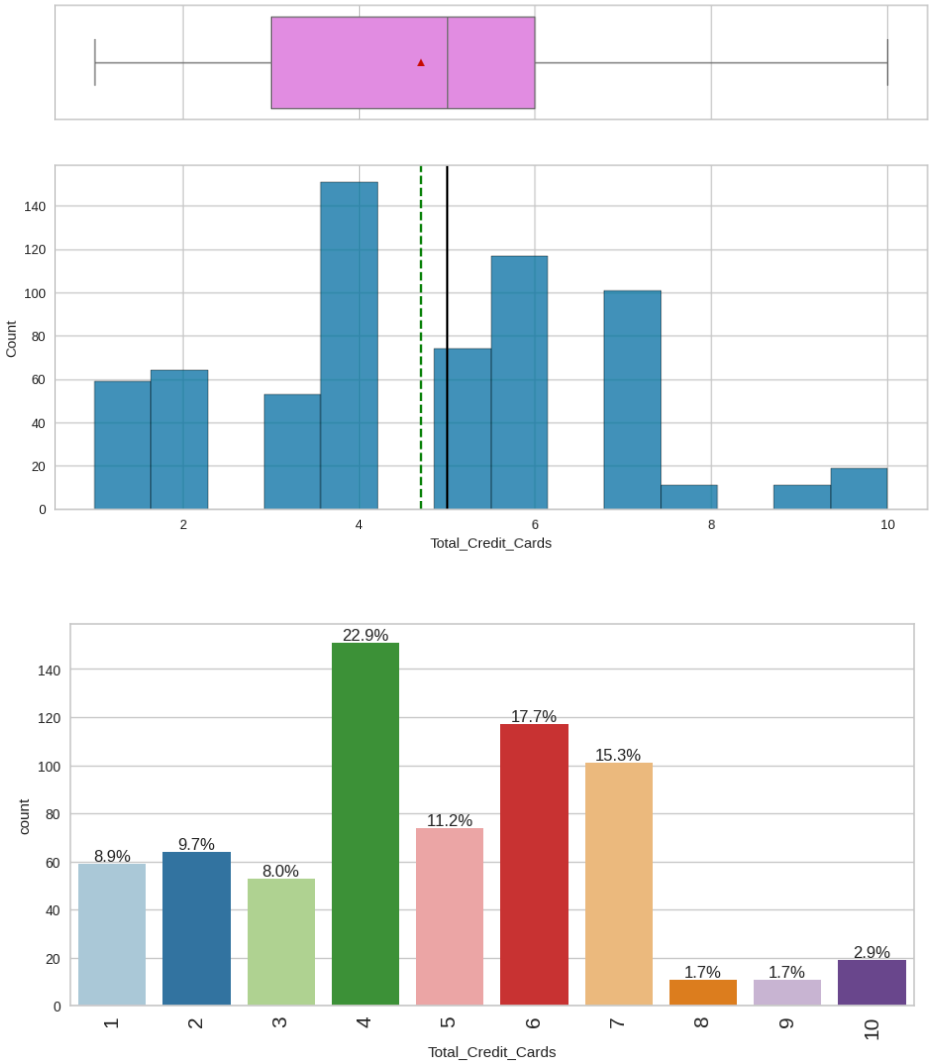

Fig 3: Observations on Total_Credit_Cards

Observations:

- Most people have 4 credit cards (about 23%).
- Many also have 6 (18%) or 7 (15%) cards.
- Fewer people have 1–3 cards.
- Very few have 8, 9, or 10 cards.
- Having 4–7 credit cards is most common, while very high or very low card counts are rare.
- The data is fairly balanced, with no extreme outliers.

1.3. Observations on Total_visits_bank

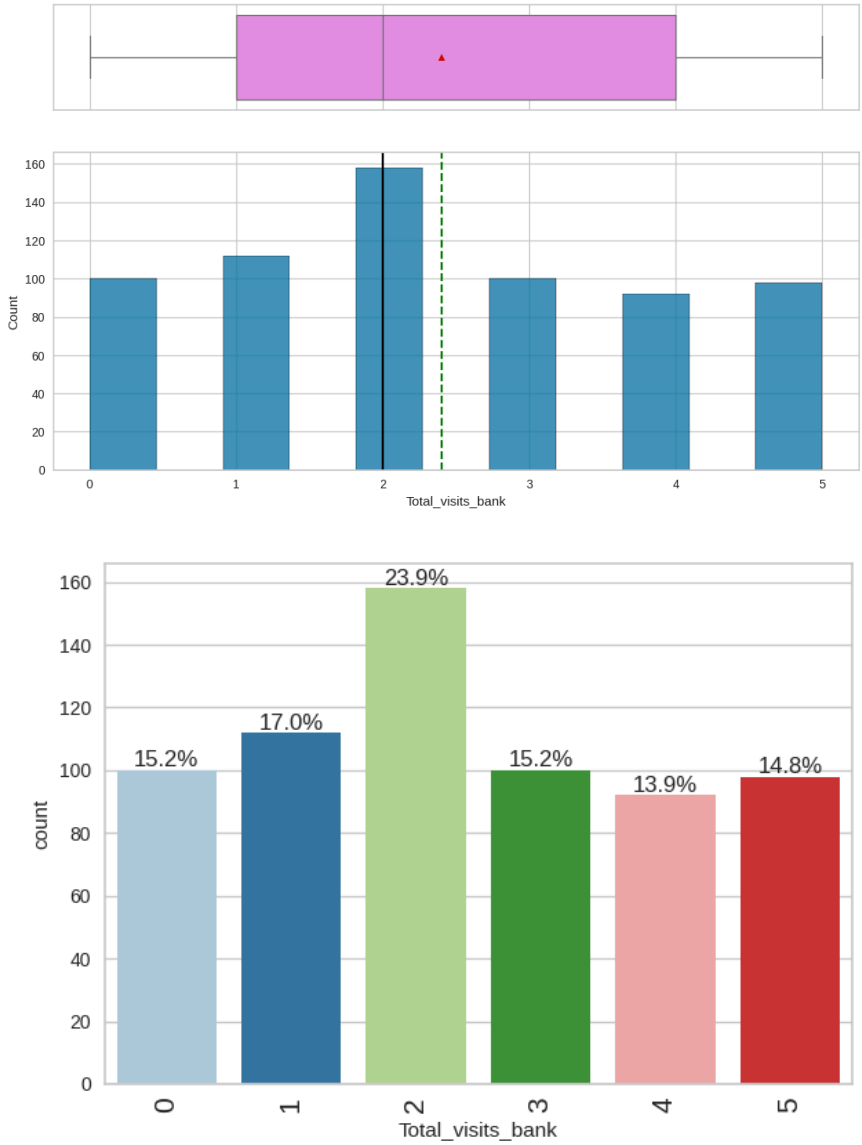

Fig 4: Observations on Total_visits_bank

Observations:

- Most customers visit the bank about 2 times(about 24%).
- Visits of 1, 3, 4, and 5 times are also common (around 14–17% each).
- Visits range from 0 to 5, so no extreme outliers.
- The distribution looks fairly spread out, with customers visiting at different levels.
- Both the mean and median are around 2 visits, showing balance.

1.4. Observations on Total_visits_online

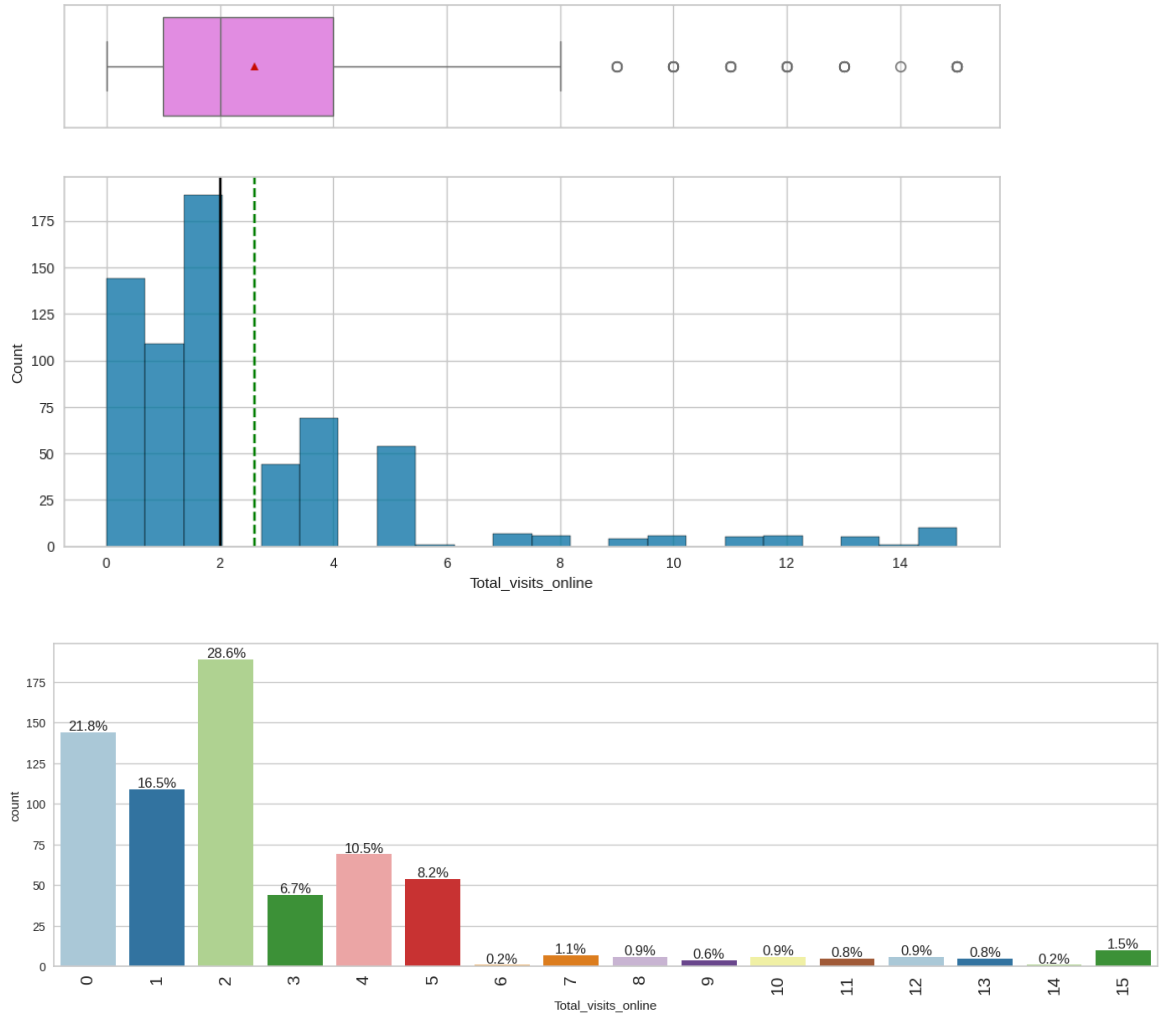

Fig 5: Observations on Total_visits_online

Observations:

- Most customers visit online 1 to 2 times(about 22% to 29%).
- Smaller groups visit 3–5 times (7–11%).
- Very few people use online services more than 6 times (less than 2% each).
- A few customers visit much more (up to 15 times), which appear as outliers.
- The data is right-skewed, meaning many low values and few very high ones.
- The majority of customers make low to moderate online visits.

1.5. Observations on Total_calls_made

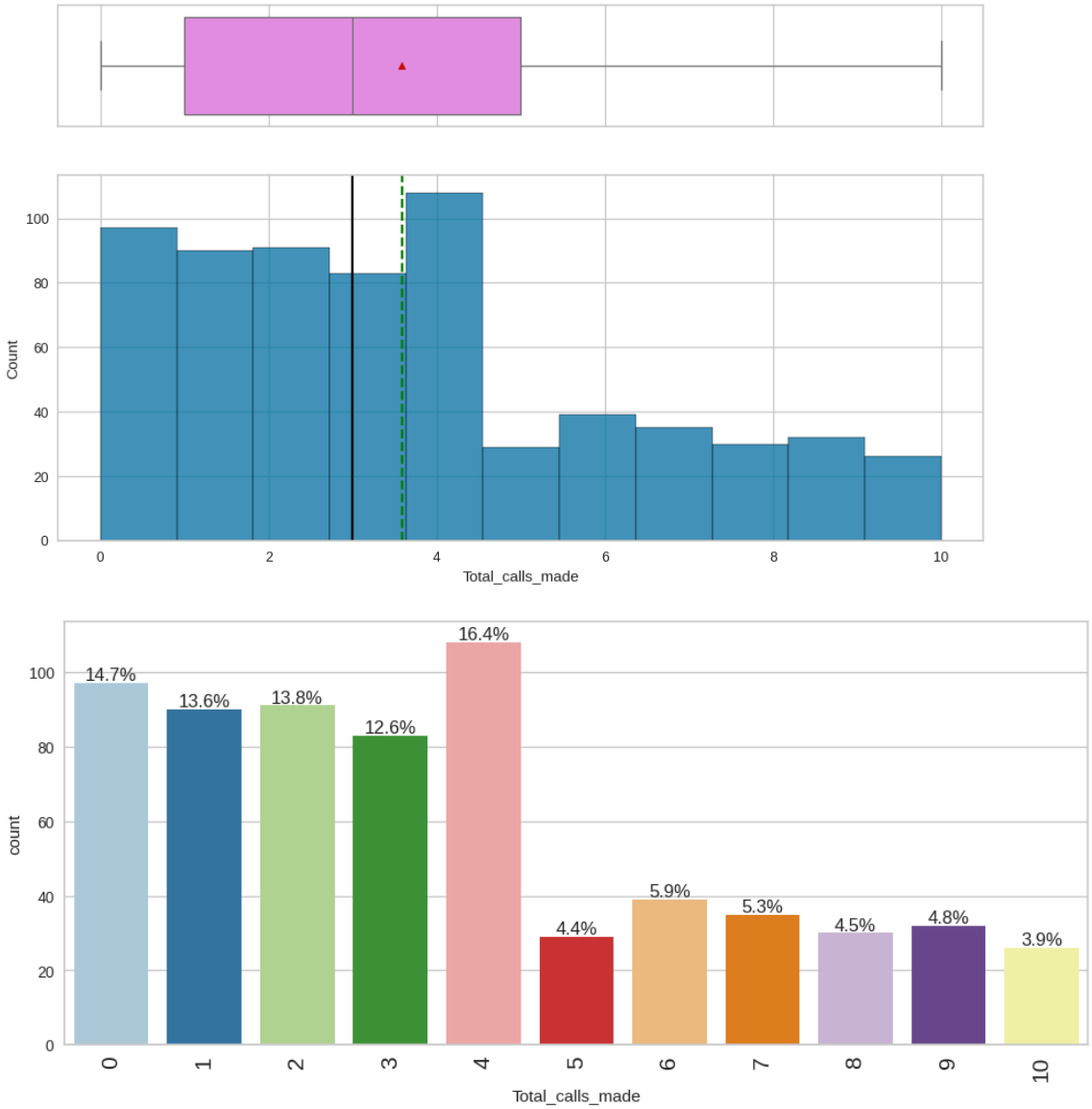

Fig 6: Observations on Total_calls_made

Observations:

- Most customers make about 4 calls (about 16%).
- Some customers make very few calls 0–3 calls (each around 13–15%).
- Fewer customers make 5 or more calls (each below 6%).
- The data is fairly spread out across the range.
- There are no extreme outliers, and calls are distributed more evenly than visits.

1.6. Plot Of Numeric Variables

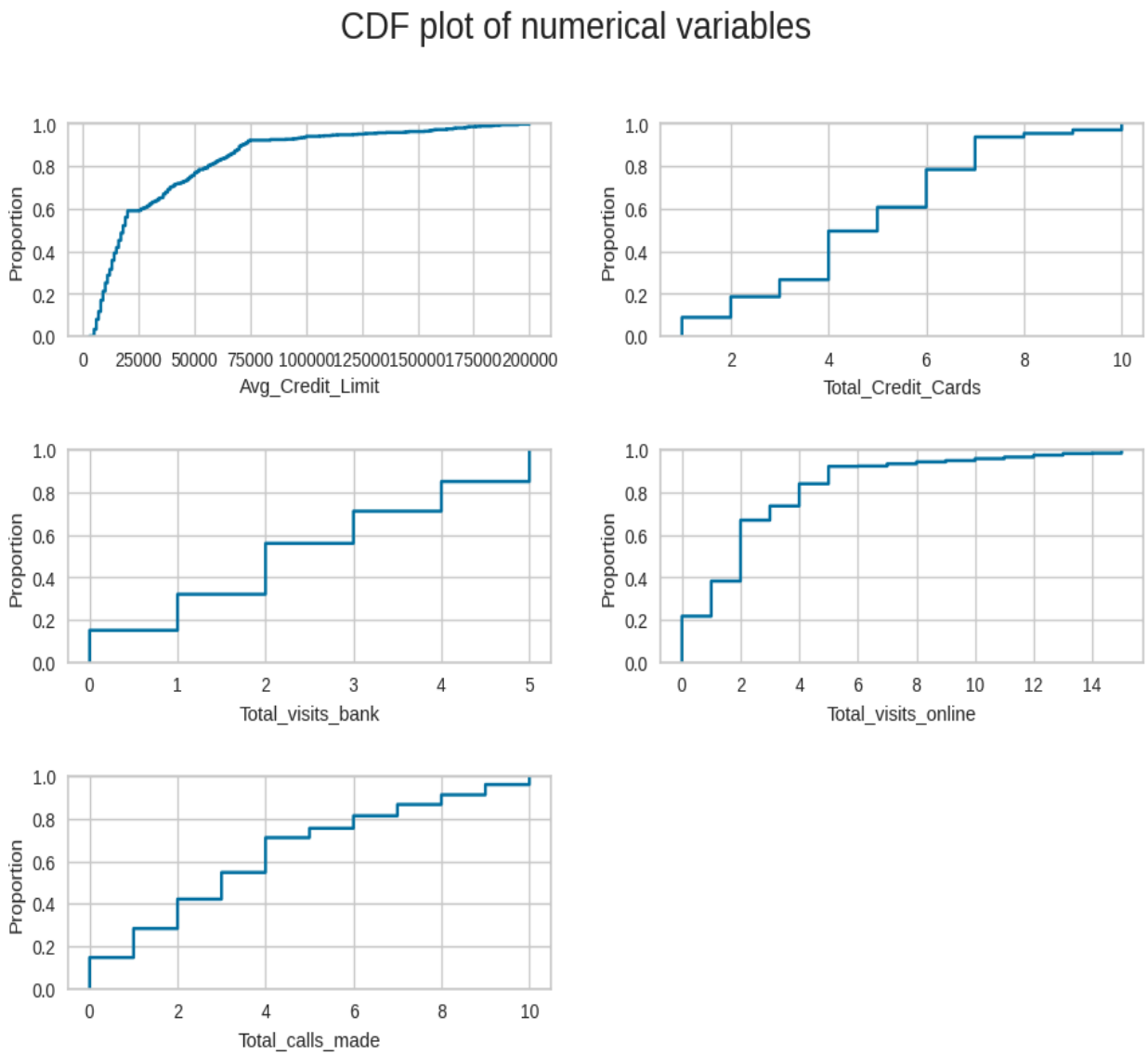

Fig 7: Plot Of Numeric Variables

Bivariant Analysis

1.7. Observation on Correlation Matrix on Numerical Variables

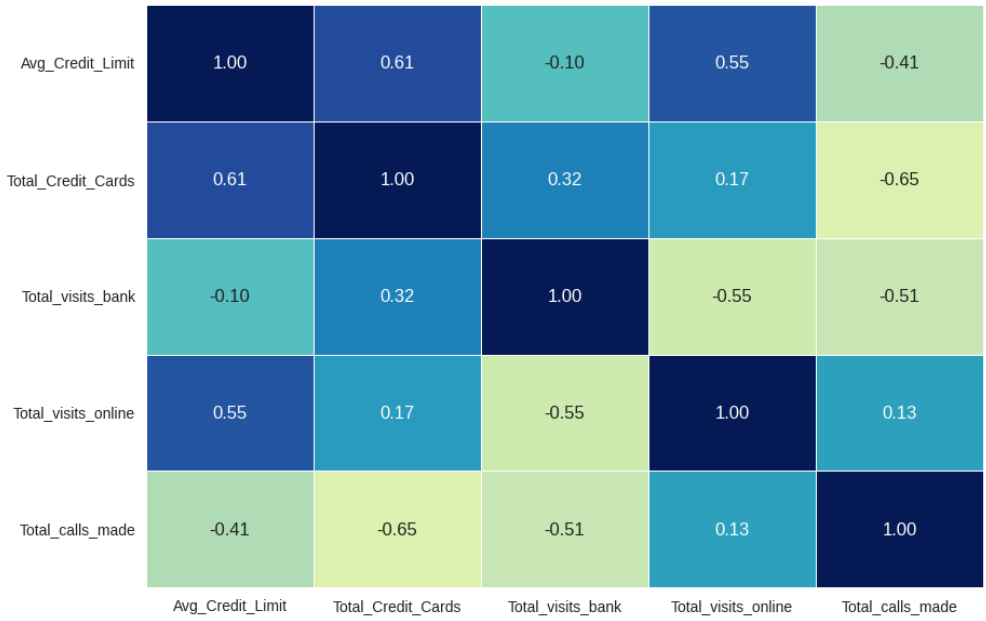

Fig 8: Observation on Correlation Matrix on Numerical Variables

Observations:

- Avg_Credit_Limit has:
  - Strong positive correlation with Total_Credit_Cards (0.61).
  - Positive correlation with Total_visits_online (0.55).
  - Negative correlation with Total_calls_made (-0.41).
- Total_Credit_Cards has:
  - Moderate positive correlation with Total_visits_bank (0.32).
  - Strong negative correlation with Total_calls_made (-0.65).
- Total_visits_bank has:
  - Strong negative correlation with Total_visits_online (-0.55).
  - Moderate negative correlation with Total_calls_made (-0.51).
- Total_visits_online has:
  - Slight positive correlation with Total_calls_made (0.13).
- Total_calls_made generally shows negative correlations with most variables (except a weak positive with online visits).

1.8. Observation on all the column using pairplot

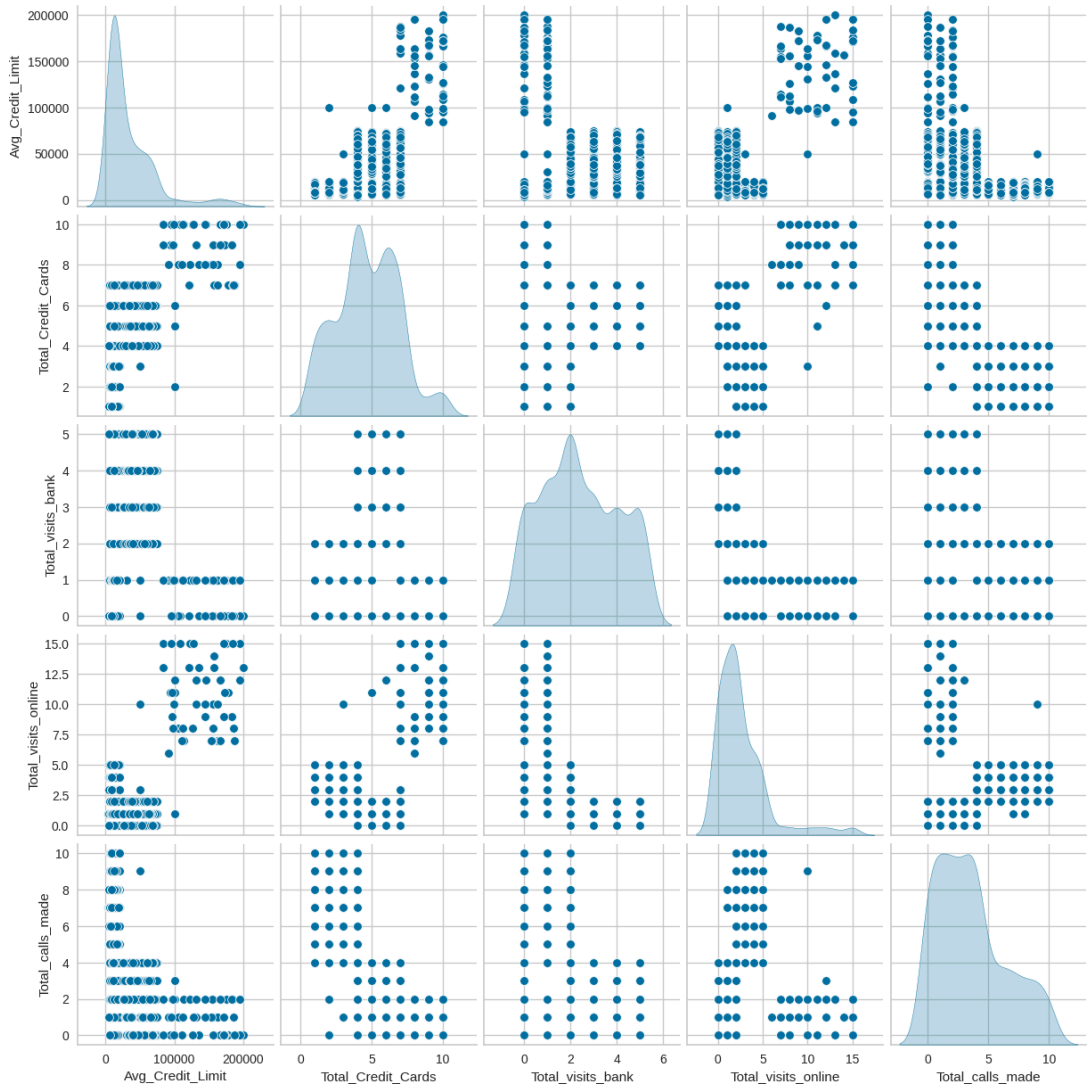

Fig 9: Observation on all the column using pairplot

Observations:

- Most customers are low usage – they have low credit limits, few cards, few visits, and few calls.
- Customers with higher credit limits and more cards prefer online banking and make fewer calls.
- Customers who visit the bank more tend to use online less and call more.
- Only a small group of customers stand out with very high credit limits, many cards, or heavy usage.

1.9. Observations Between Total_Credit_Cards and Avg_Credit_Limit

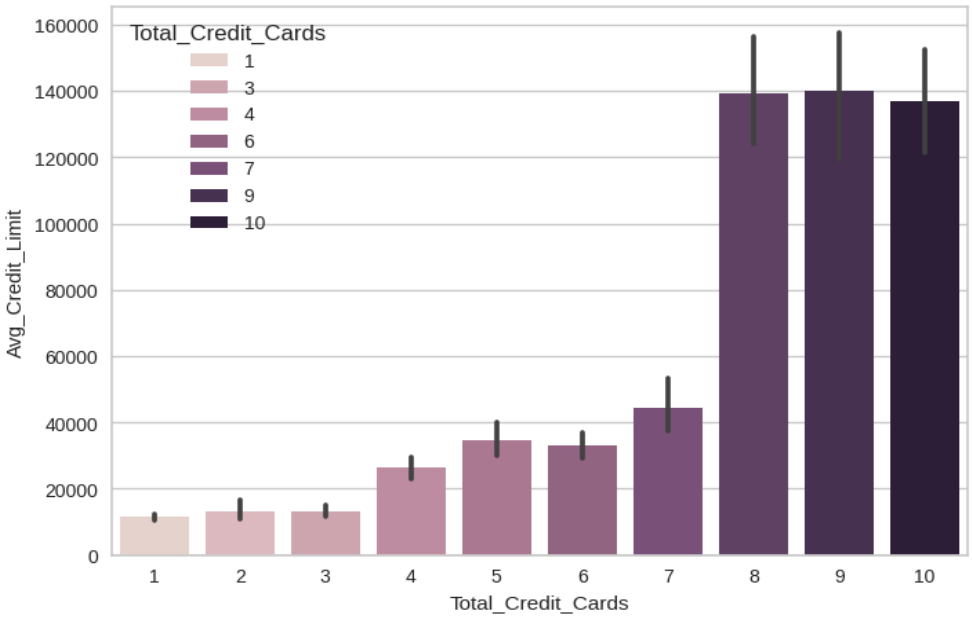

Fig 10: Observations Between Total_Credit_Cards and Avg_Credit_Limit

Observations:

- Having just one or two cards means a small credit limit.
- The more credit cards a person has, the bigger their total limit gets.
- After about seven cards, the average credit limit jumps up by a lot.
- People with the most cards have the most credit to spend, but the amount can vary a lot between different people.

1.10. Observations on Total_Credit_Cards and Total_calls_made

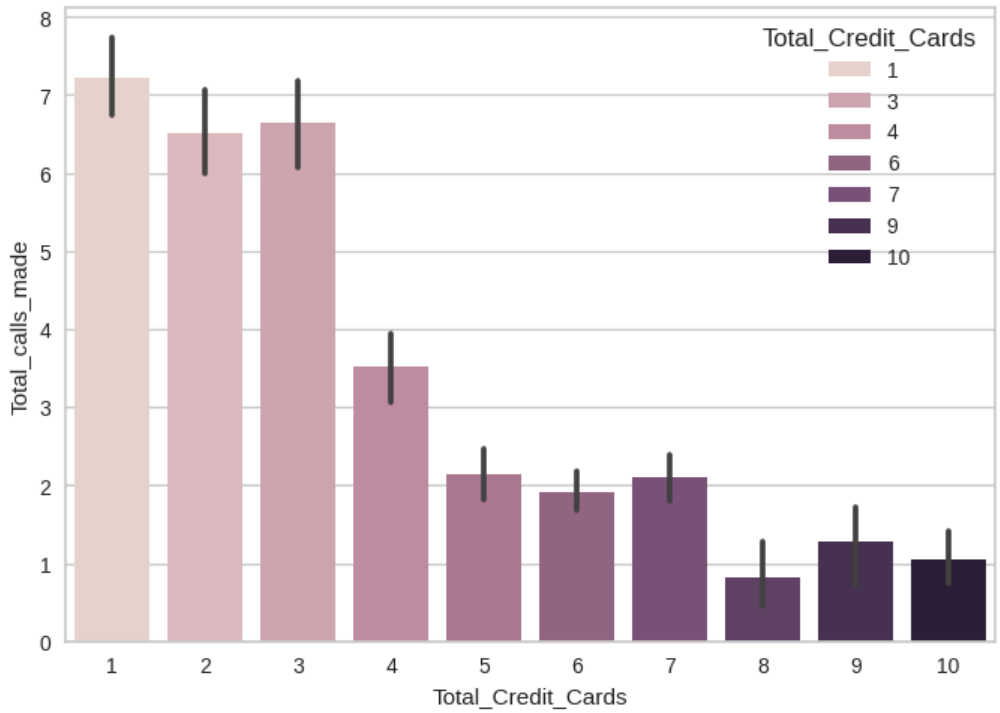

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Fig 11: Observations on Total_Credit_Cards and Total_calls_made

Observations:

- Most customers made between 0 and 3 calls (majority of the data is concentrated there).
- Very few customers made more than 5 calls.
- The distribution is right-skewed, meaning higher call counts are rare compared to lower ones.

Overall, the majority of customers contact only a few times, while a small group contacts very frequently.

1.11. Observations on Total_Credit_Cards and Total_visits_bank

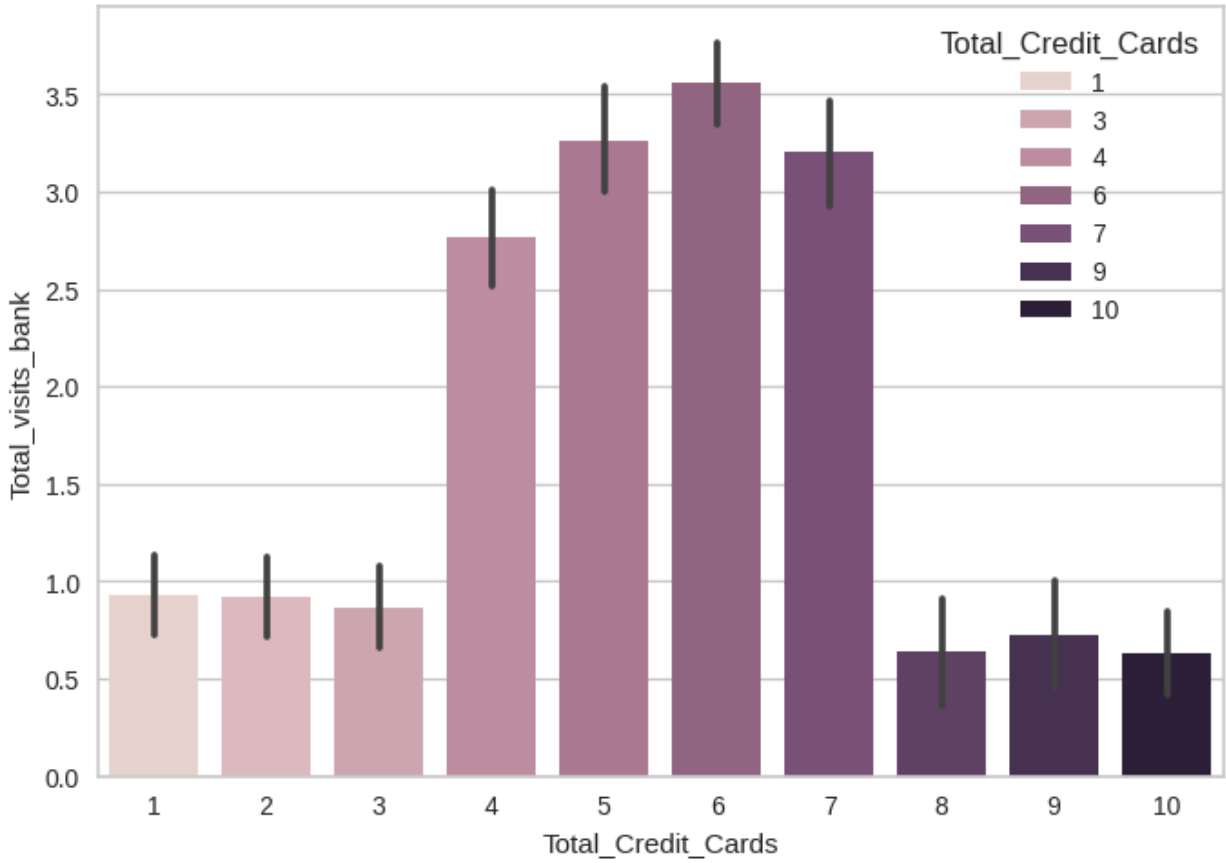


Fig 12: Observations on Total_Credit_Cards and Total_visits_bank

Observations:

- Customers with 4–7 credit cards make the highest number of bank visits.
- Customers with very few cards (1–2) or many cards (9–10) visit the bank less often.
- The distribution looks concentrated in the mid-range (4–7 cards), showing that moderate cardholders interact with the bank more frequently.
- Overall, customers with a balanced number of credit cards tend to engage with the bank more than those at the extremes.

1.12. Observations on Total_Credit_Cards and Total_visits_online

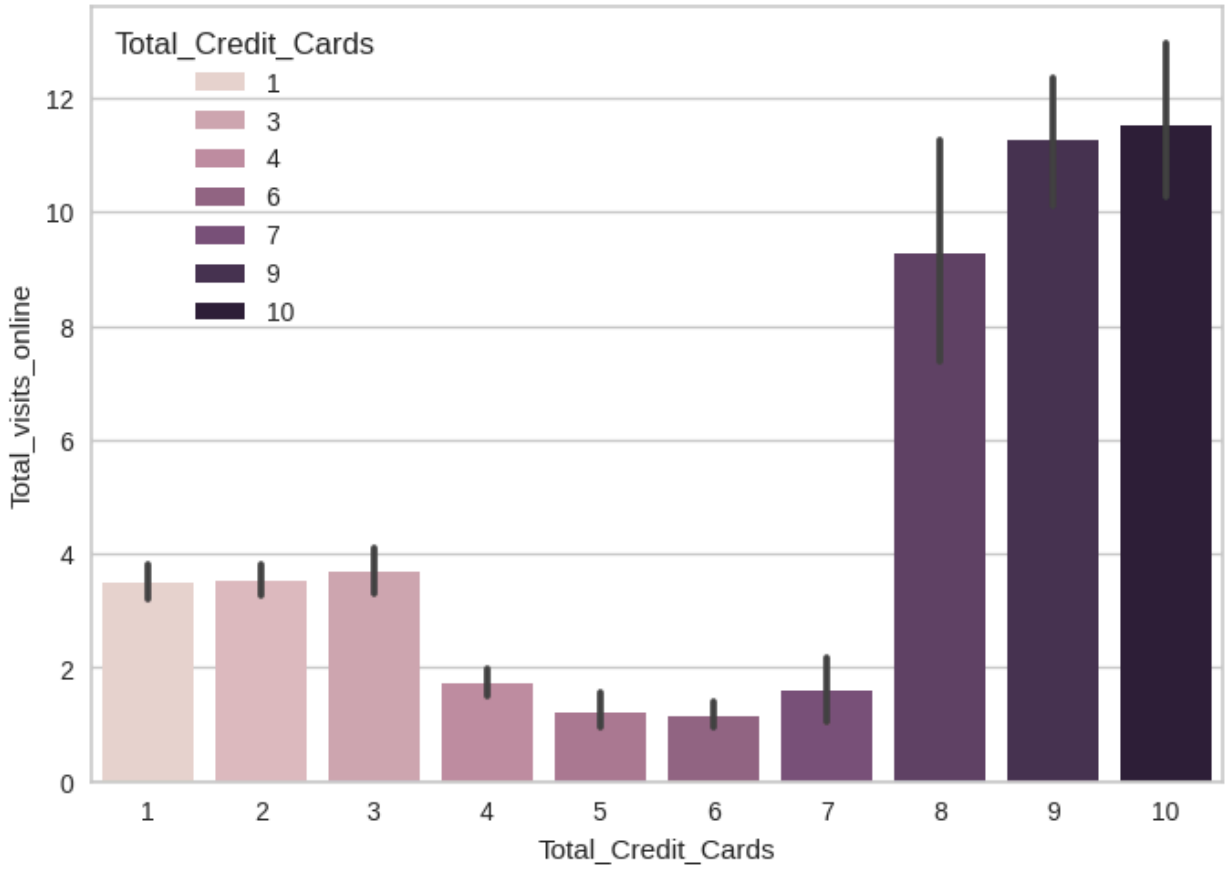


Fig 13: Observations on Total_Credit_Cards and Total_visits_online

Observations:

- Customers with a higher number of credit cards (8–10) make the most online visits.
- Customers with fewer cards (1–3) rarely use online channels.
- Overall, more credit card users rely more on online platforms compared to customers with fewer cards.

1.13. Observations on Total_visits_bank and Total_calls_made

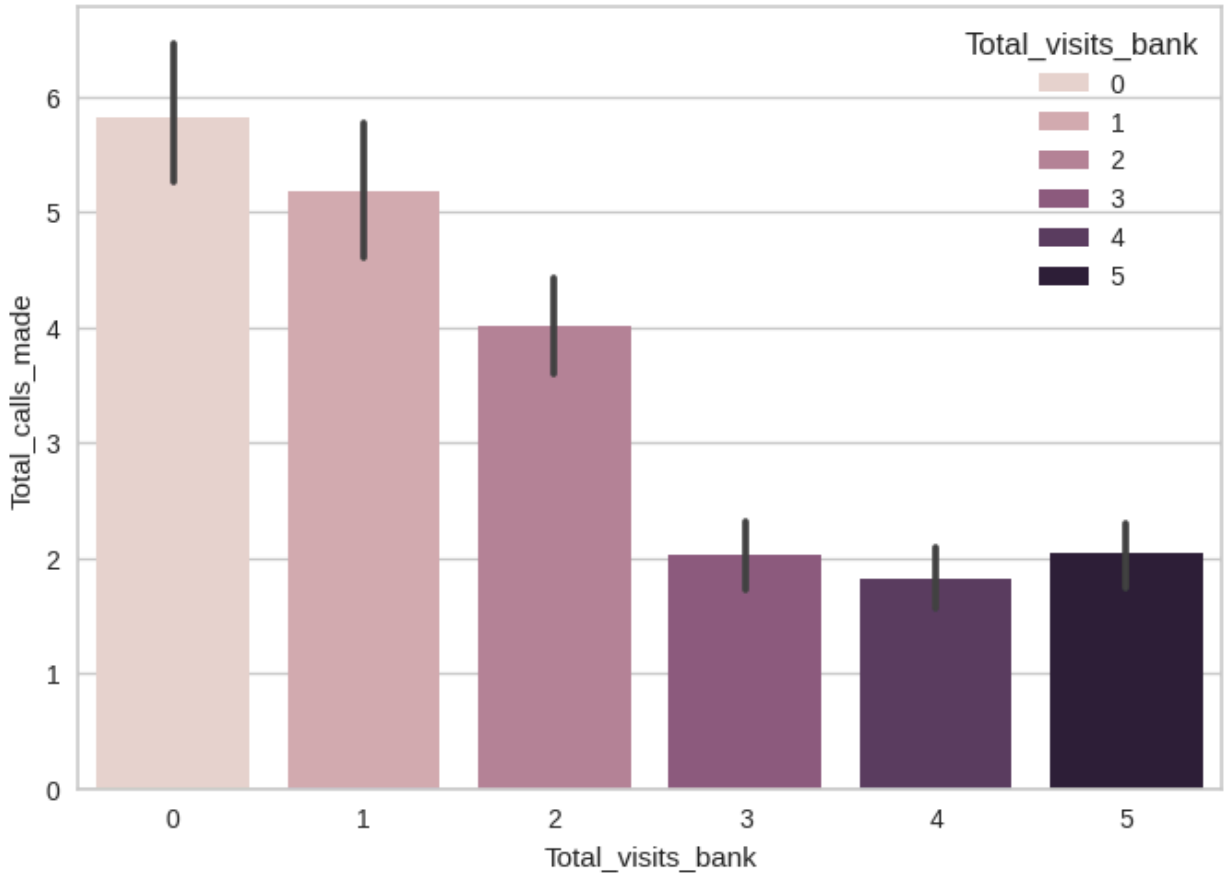


Fig 14: Observations on Total_visits_bank and Total_calls_made

Observations:

- Customers with 0–2 bank visits tend to make more calls compared to those with higher visits.
- As bank visits increase (3–5), the number of calls made generally decreases.
- This shows a negative relationship – customers who visit the bank more often usually make fewer calls.

1.14. Observations on Total_visits_online and Total_calls_made

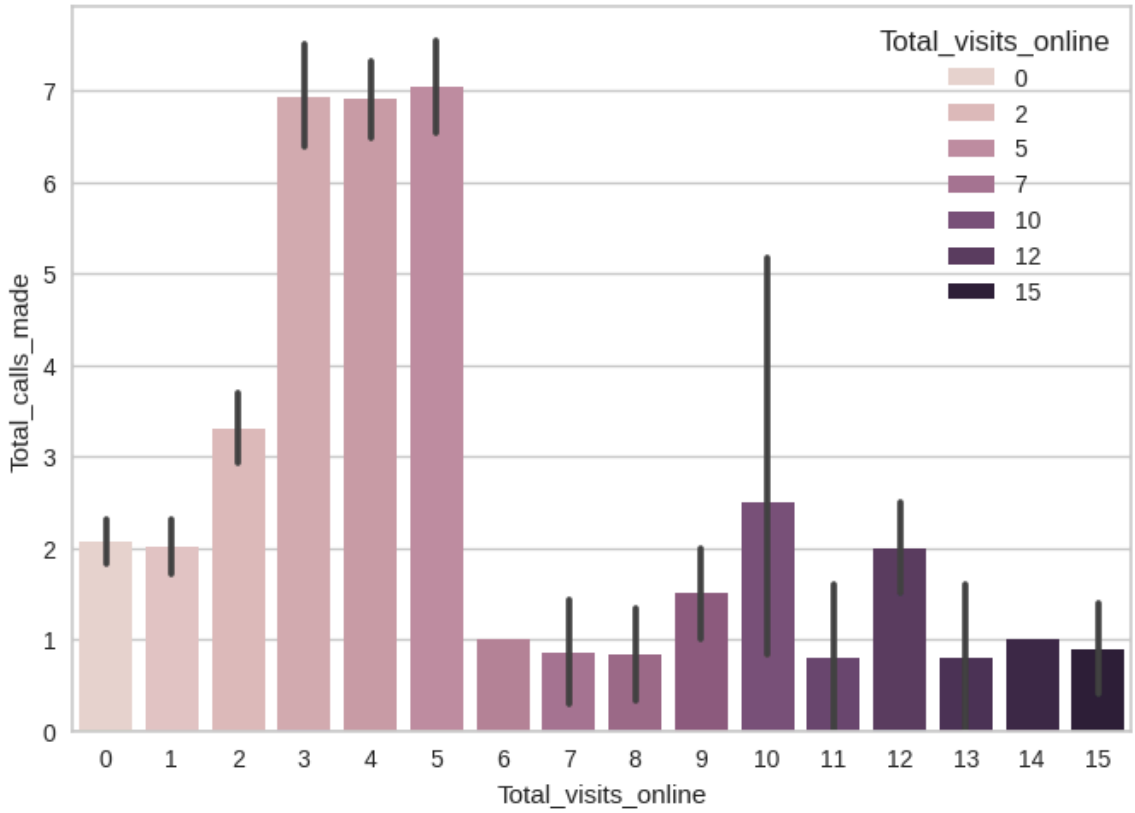


Fig 15: Observations on Total_visits_online and Total_calls_made

Observations:

- Customers with moderate online visits (3–6 times) tend to make the highest number of calls.
- Customers with very high online visits (10–15 times) make fewer calls, suggesting they rely more on digital channels.
- Customers with low online visits (0–2 times) also keep their calls relatively low.

Data Processing

Data Preprocessing

- Missing value treatment (if needed)
- Outlier Detection and Treatment (with rationale if needed)
- Feature Engineering (with rationale if needed)
- Data Scaling (with rationale if needed)

Missing value treatment

Observations:

- There are no missing values in the given dataset.

	0
Avg_Credit_Limit	0
Total_Credit_Cards	0
Total_visits_bank	0
Total_visits_online	0
Total_calls_made	0

dtype: int64

Fig 16: Checking Missing Values

Outlier detection and treatment

Detecting Outliers

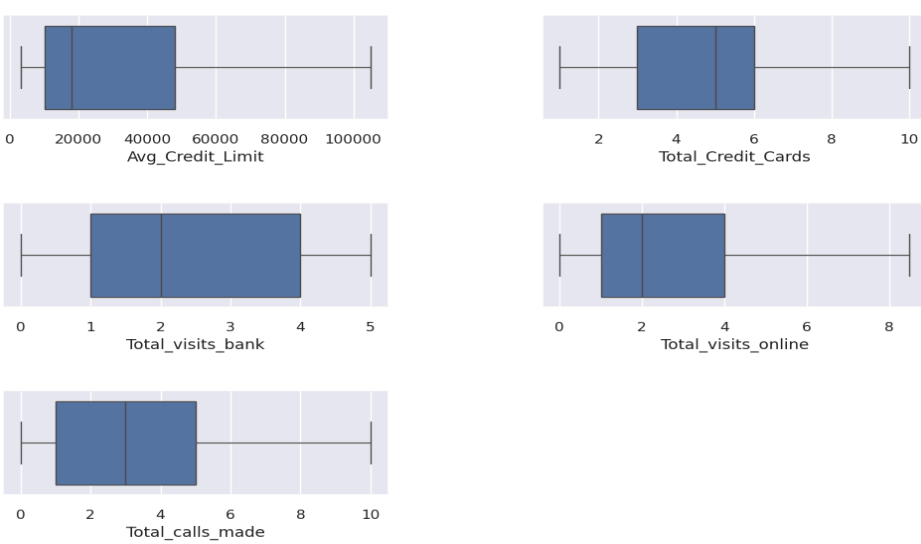


Fig 17: Detecting Outliers

Observations:

- Total_Credit_Crards, Total_visit_bank and Total_calls_made columns have no outliers.
- Avg_Credit_Limit and Total_visits_online columns have outliers.
- We need to treat these outliers.

Treating Outliers

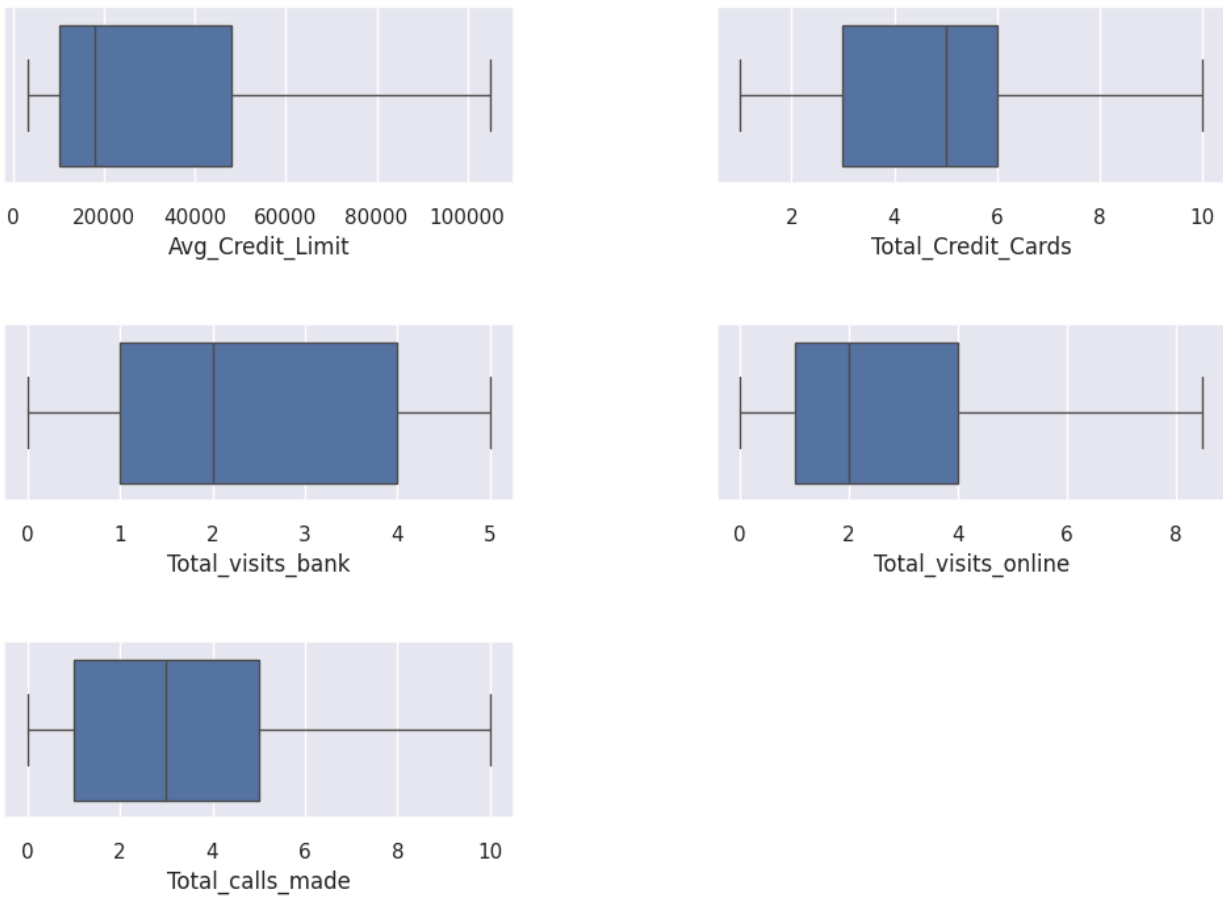


Fig 18: Treating Outliers

Observations:

- The outliers have been treated completely.

Scaling The Data

- Scaling the dataset before clustering using StandardScaler().
- Created dataframe copies for k-means and hierarchical clustering.

Model Building

K-means Clustering

Checking Elbow Plot

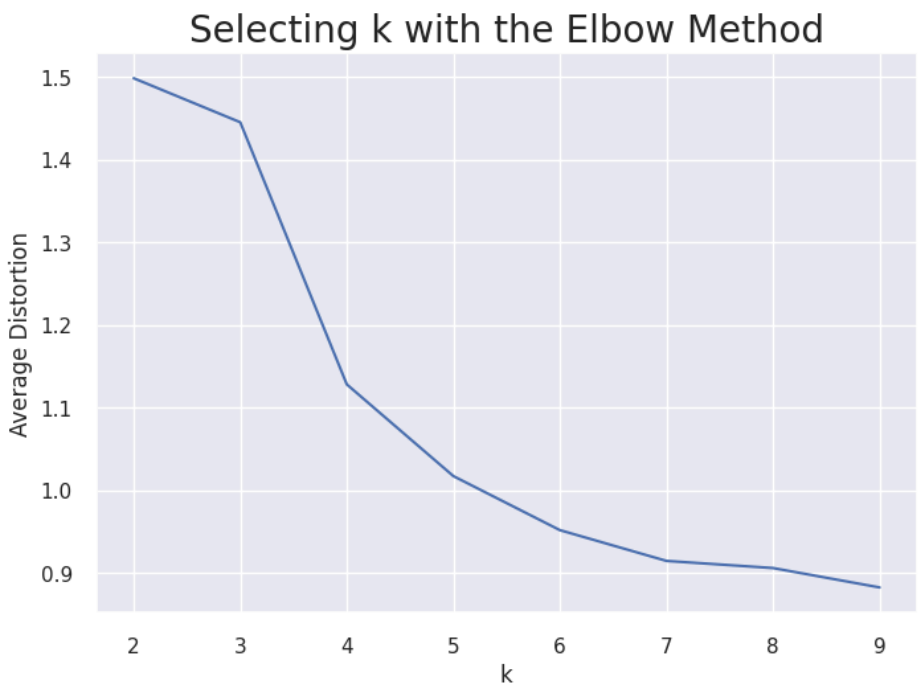


Fig 19: Elbow Plot

Fig 20: Graph to determine the optimal value of k

The elbow method indicates that the optimal number of clusters for KMeans is **k = 4**, as the distortion score begins to level off beyond this point.

Checking Silhouette Scores

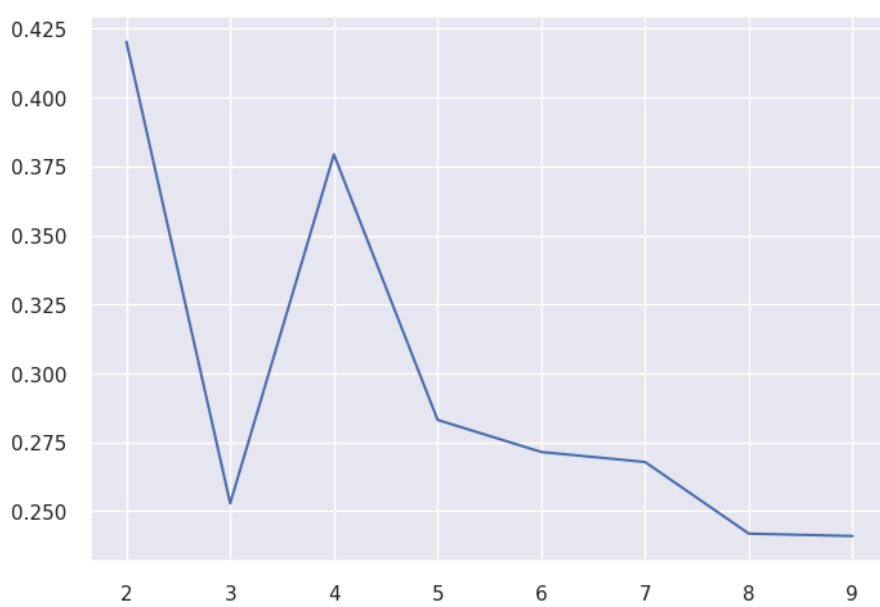


Fig 21: Graph To Check Silhouette Scores

Fig 22: Silhouette Score Elbow For KMeans Clustering

From the silhouette scores, it seems that 4 is a good value for k.

Silhouette Plot

1.15. Silhouette Plot of KMeans Clustering in 2 Centers

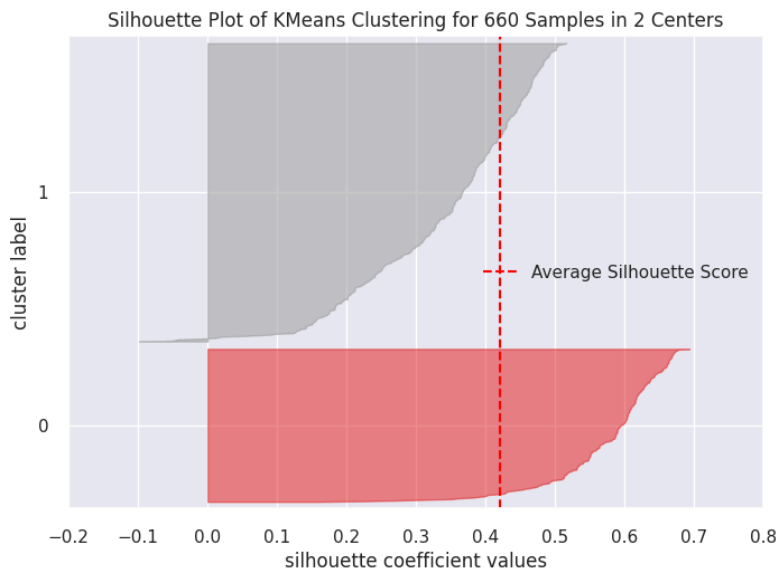


Fig 23: Silhouette Plot of KMeans Clustering in 2 Centers

1.16. Silhouette Plot of KMeans Clustering in 3 Centers

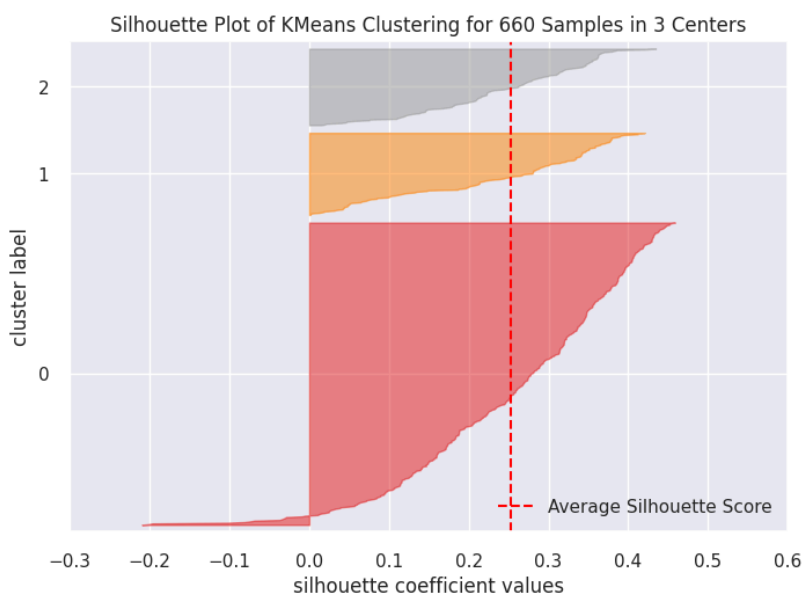


Fig 24: Silhouette Plot of KMeans Clustering in 3 Centers

1.17. Silhouette Plot of KMeans Clustering in 4 Centers

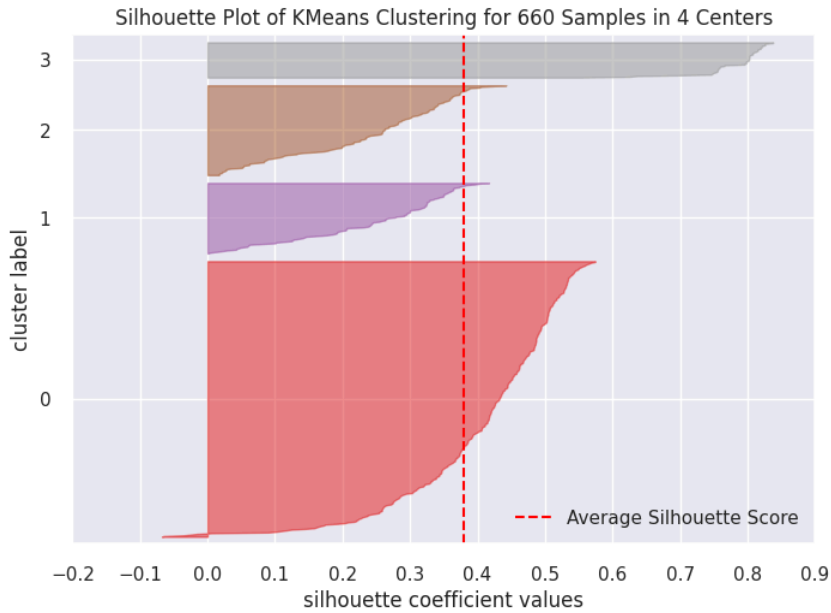


Fig 25: Silhouette Plot of KMeans Clustering in 4 Centers

1.18. Silhouette Plot of KMeans Clustering in 5 Centers

Fig 26: Silhouette Plot of KMeans Clustering in 5 Centers

1.19. Silhouette Plot of KMeans Clustering in 6 Centers

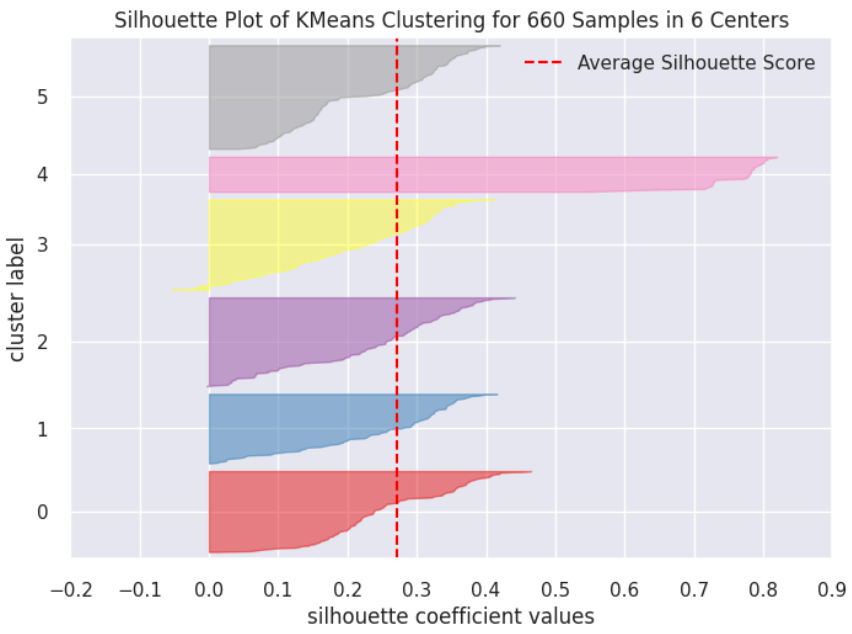


Fig 27: Silhouette Plot of KMeans Clustering in 6 Centers

1.20. Silhouette Plot of KMeans Clustering in 7 Centers

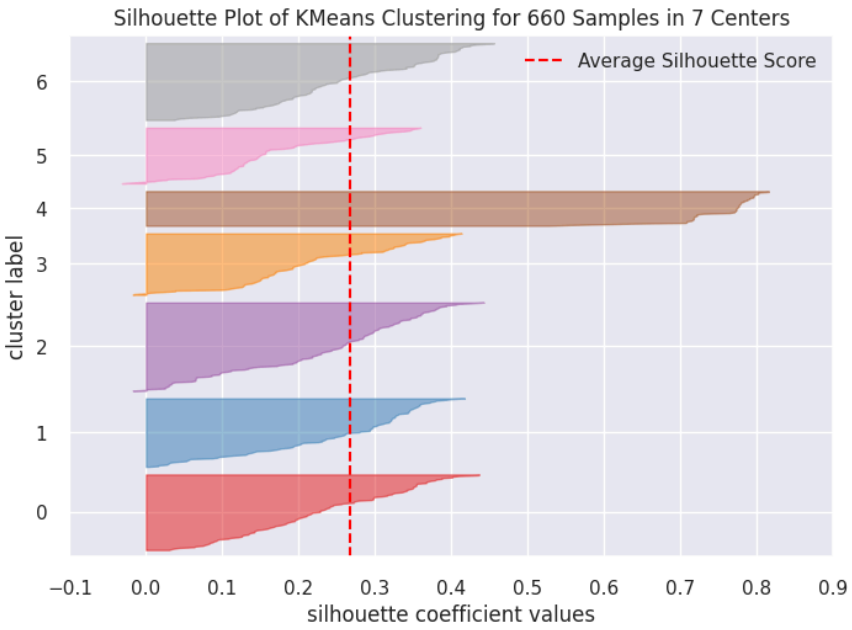


Fig 28: Silhouette Plot of KMeans Clustering in 7 Centers

Creating Final Model

- We will build our model using the KMeans().

```
KMeans
KMeans(n_clusters=4, random_state=1)
```

Fig 29: Final Model of KMeans Clustering

- Adding KMeans cluster labels to the KMeans dataframe(km_df)

Hierarchical Clustering

Computing Cophenetic Correlation

```
Cophenetic correlation for Euclidean distance and single linkage is 0.6666982080229052.
Cophenetic correlation for Euclidean distance and complete linkage is 0.8207884321476555.
Cophenetic correlation for Euclidean distance and average linkage is 0.8926672966587861.
Cophenetic correlation for Euclidean distance and weighted linkage is 0.8872365340578665.
Cophenetic correlation for Chebyshev distance and single linkage is 0.6882876568342585.
Cophenetic correlation for Chebyshev distance and complete linkage is 0.7672199195985938.
Cophenetic correlation for Chebyshev distance and average linkage is 0.8907216945489854.
Cophenetic correlation for Chebyshev distance and weighted linkage is 0.782728724334691.
Cophenetic correlation for Mahalanobis distance and single linkage is 0.658393428048562.
Cophenetic correlation for Mahalanobis distance and complete linkage is 0.388494379210295.
Cophenetic correlation for Mahalanobis distance and average linkage is 0.7998609088477292.
Cophenetic correlation for Mahalanobis distance and weighted linkage is 0.6221434055030449.
Cophenetic correlation for Cityblock distance and single linkage is 0.6711679581499276.
Cophenetic correlation for Cityblock distance and complete linkage is 0.8365243185904129.
Cophenetic correlation for Cityblock distance and average linkage is 0.8905302338528754.
Cophenetic correlation for Cityblock distance and weighted linkage is 0.8419484433810667.
```

Fig 30: Cophenetic Correlation

Observation:

It is found that, highest cophenetic correlation is 0.8926672966587861, which is obtained with Euclidean distance and average linkage.

Dendrograms For Different Linkage Methods.

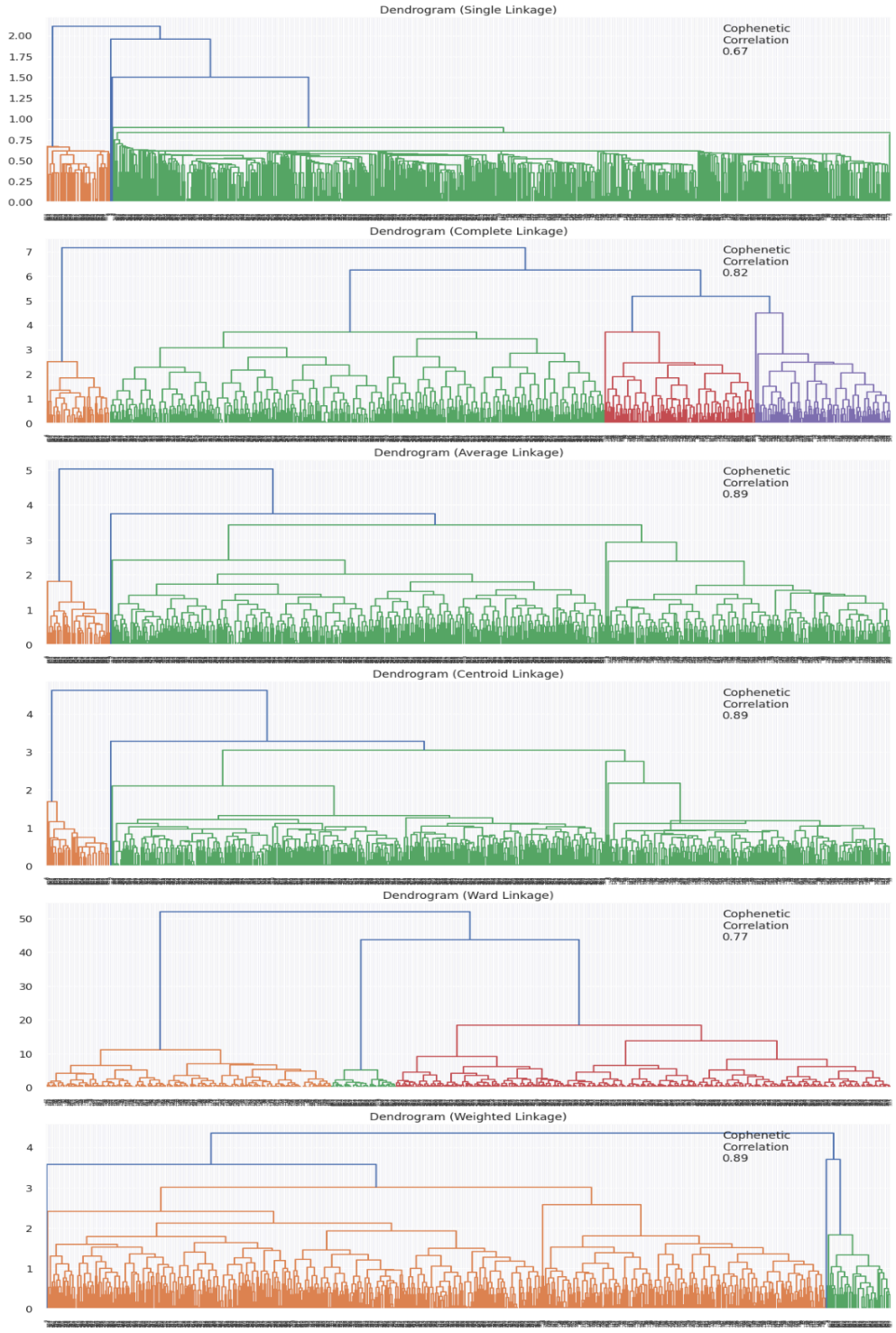


Fig 31: Dendrograms For Different Linkage Methods

Observations:

- Single linkage has the lowest correlation (0.67), so it is not very good.
- Average, Centroid, and Weighted linkage give the highest correlation (0.89).
- Ward linkage has only moderate correlation (0.77) seems to result in the best separation between clusters, even though its cophenetic correlation is lower than the other linkages..
- Complete linkage (0.82) performs better than single but not the best.
- Overall, Average, Centroid, and Weighted are the most reliable choices here.

Creating Final Model

- We will build our model using the AgglomerativeClustering() using 4 clusters, elucidian metric and average linkage.
- Adding Agglomerative cluster labels to the KMeansdataframe(hc_df).

Cluster Profiling and Comparison

Cluster Profiling: K-means Clustering

- Grouping Dataframe Based On KMeans Cluster Labels.

KM_segments	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	33851.948052	5.516883	3.493506	0.979221	1.994805
1	12464.646465	2.292929	0.888889	3.813131	8.737374
2	11904.761905	2.523810	0.976190	3.325397	5.380952
3	102660.000000	8.740000	0.600000	8.180000	1.080000

Table 5: Grouping Dataframe Based On KMeans Cluster Labels

- Adding “Freq” as a new column to Dataframe.

KM_segments	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Freq
0	33851.948052	5.516883	3.493506	0.979221	1.994805	385
1	12464.646465	2.292929	0.888889	3.813131	8.737374	99
2	11904.761905	2.523810	0.976190	3.325397	5.380952	126
3	102660.000000	8.740000	0.600000	8.180000	1.080000	50

Table 6: Adding “Freq” as a new column to Dataframe

- Highlighted Maximum Value In Each Column.

KM_segments	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Freq
0	33851.948052	5.516883	3.493506	0.979221	1.994805	385
1	12464.646465	2.292929	0.888889	3.813131	8.737374	99
2	11904.761905	2.523810	0.976190	3.325397	5.380952	126
3	102660.000000	8.740000	0.600000	8.180000	1.080000	50

Table 7: Highlighted Maximum Value In Each Column

Grouping Based On Avg_Credit_Limit

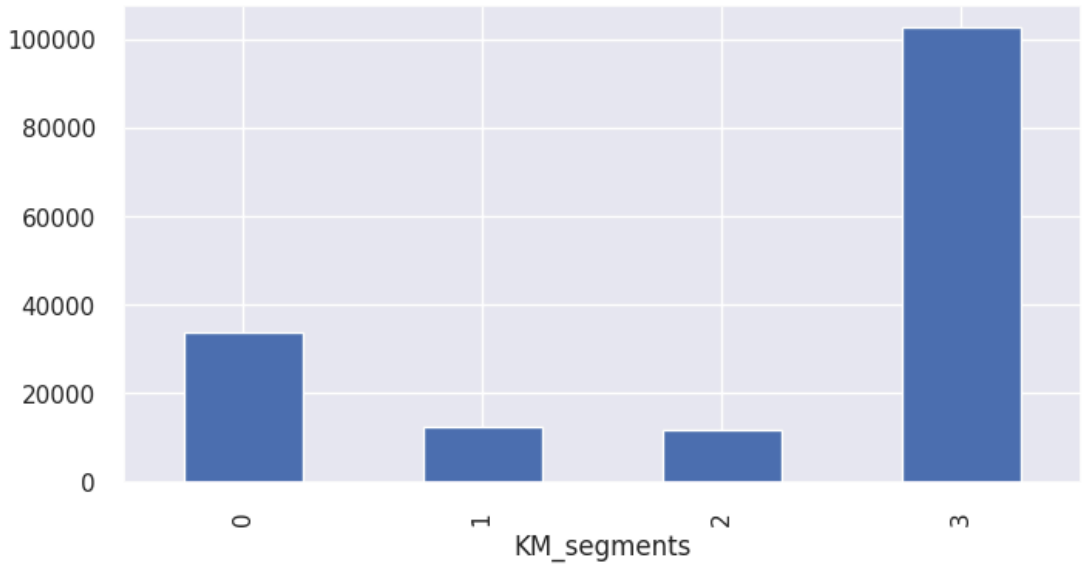


Fig 32: Grouping Based On Avg_Credit_Limit

The bar chart shows that Cluster 3 has the largest number of data points, while Clusters 1 and 2 have the smallest.

Graph Of Numerical Variables For Each Cluster

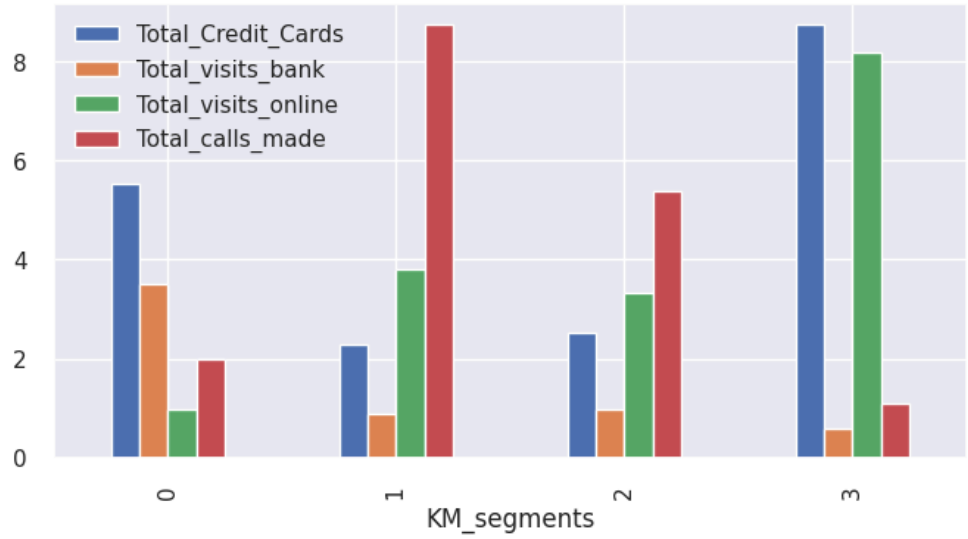


Fig 33: Graph Of Numerical Variables For Each Cluster

Boxplot Of Numerical Variables For Each Cluster

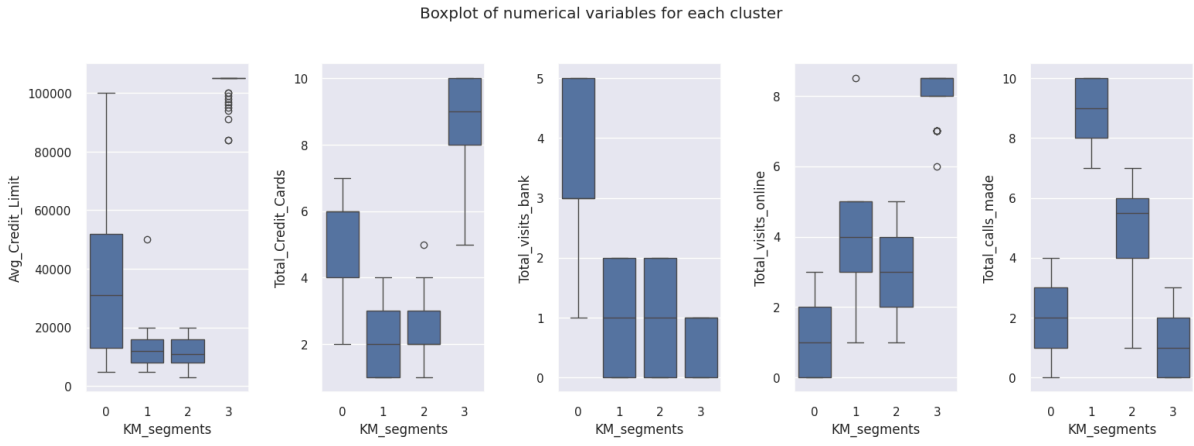


Fig 34: Boxplot of numerical variables for each cluster

Insights On KMeans Clustering

- Cluster 0 (High Credit & High Engagement)
  - Customers in this group have higher credit limits, multiple credit cards, and frequent visits to both the bank and online channels.
  - They represent the most valuable customer segment, highly engaged with multiple touchpoints.
- Cluster 1 (Low Visits, Limited Usage)
  - Customers here hold fewer credit cards and make very few bank or online visits.
  - Engagement is minimal, suggesting low product usage and limited profitability.
- Cluster 2 (Moderate Users)
  - This group has average credit limits and card usage.
  - They show some engagement across online and offline channels but not at the level of Cluster 0.
  - They represent a growth potential segment if nurtured properly.
- Cluster 3 (Very Low Engagement / At-Risk)
  - Customers in this cluster show nearly no bank or online visits and minimal card activity.
  - They are likely inactive or at risk of churn.

Cluster Profiling: Hierarchical Clustering

- Grouping Dataframe Based On Hierarchical Cluster Labels.

HC_segments	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	20666.666667	5.474510	3.678431	1.054902	2.054902
1	12197.309417	2.403587	0.928251	3.553812	6.883408
2	102660.000000	8.740000	0.600000	8.180000	1.080000
3	58916.666667	5.583333	3.113636	0.848485	1.909091

Table 8: Grouping Dataframe Based On Hierarchical Cluster Labels

- Adding “Freq” as a new column to Dataframe.

HC_segments	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Freq
0	20666.666667	5.474510	3.678431	1.054902	2.054902	255
1	12197.309417	2.403587	0.928251	3.553812	6.883408	223
2	102660.000000	8.740000	0.600000	8.180000	1.080000	50
3	58916.666667	5.583333	3.113636	0.848485	1.909091	132

Table 9: Adding “Freq” as a new column to Dataframe

- Highlighted Maximum Value In Each Column.

HC_segments	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Freq
0	20666.666667	5.474510	3.678431	1.054902	2.054902	255
1	12197.309417	2.403587	0.928251	3.553812	6.883408	223
2	102660.000000	8.740000	0.600000	8.180000	1.080000	50
3	58916.666667	5.583333	3.113636	0.848485	1.909091	132

Table 10: Highlighted Maximum Value In Each Column

Grouping Based On Avg_Credit_Limit

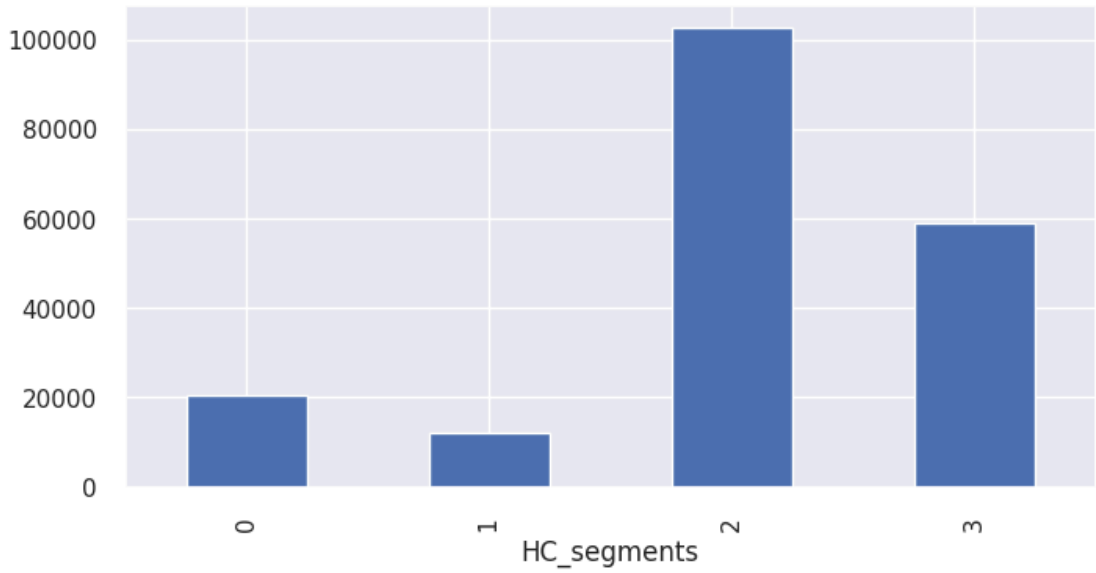


Fig 35: Grouping Based On Avg_Credit_Limit

The bar chart shows that Cluster 2 has the largest number of data points, while Clusters 1 and 3 have the smallest.

Graph Of Numerical Variables For Each Cluster

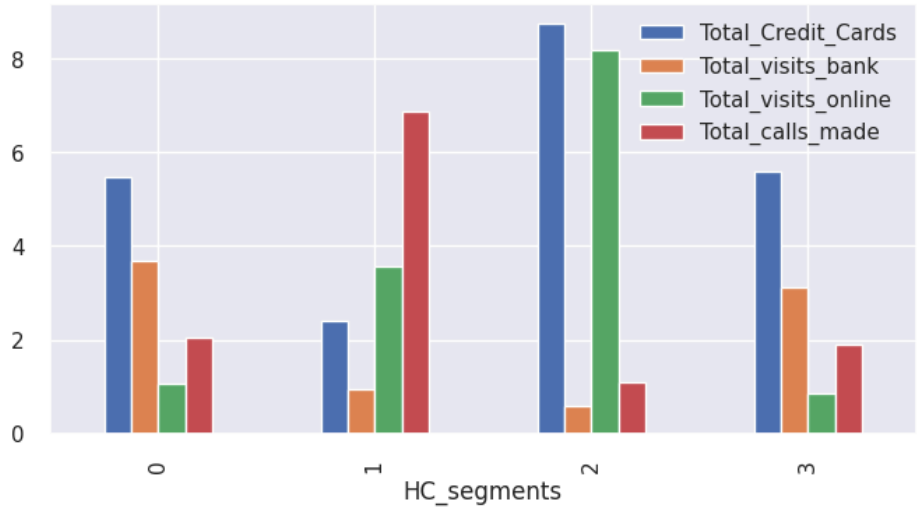


Fig 36: Graph Of Numerical Variables For Each Cluster

Boxplot Of Numerical Variables For Each Cluster

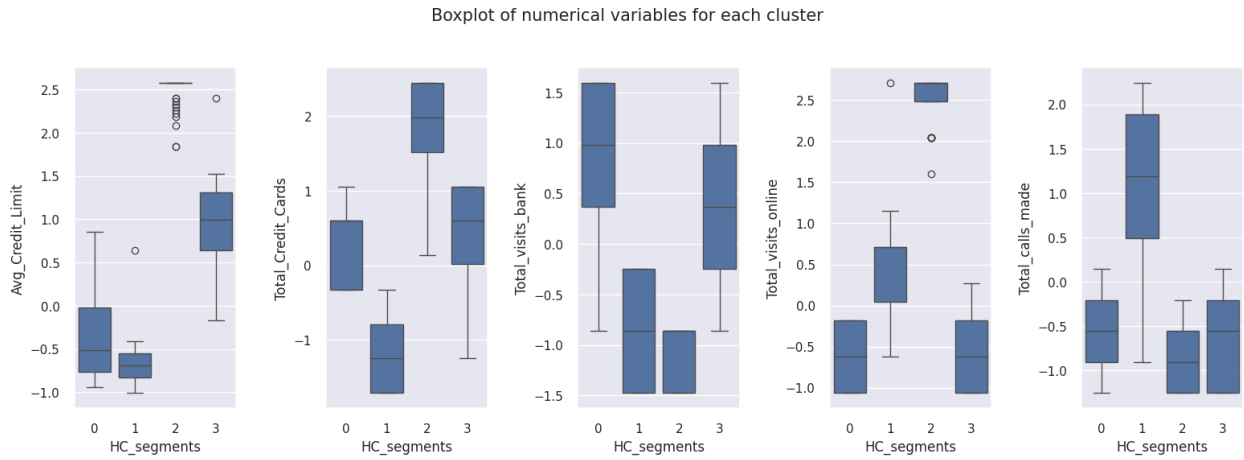


Fig 37: Boxplot Of Numerical Variables For Each Cluster

Insights On Hierarchical Clustering

- Cluster 0 — Premium & Highly Engaged
  - Customers with high credit limits, multiple credit cards, and frequent bank + online visits.
  - They form the core profitable segment, similar to the K-Means high-value cluster.
- Cluster 1 — Low Usage, Low Engagement
  - Customers with low to moderate credit limits, very few cards, and minimal online/bank interactions.
  - Indicates a group with limited product adoption.
- Cluster 2 — Balanced / Average Users
  - Customers with moderate credit limits and usage patterns.
  - Somewhat engaged, but not as active as Cluster 0.
  - Represents a potential growth segment if provided with incentives.
- Cluster 3 — Inactive / At-Risk Customers
  - Customers with almost no visits to bank or online platforms, very low card activity.
  - Likely inactive or at risk of churn, requiring reactivation strategies.

Compare cluster K-means clusters and Hierarchical clusters

KMeans Clusters

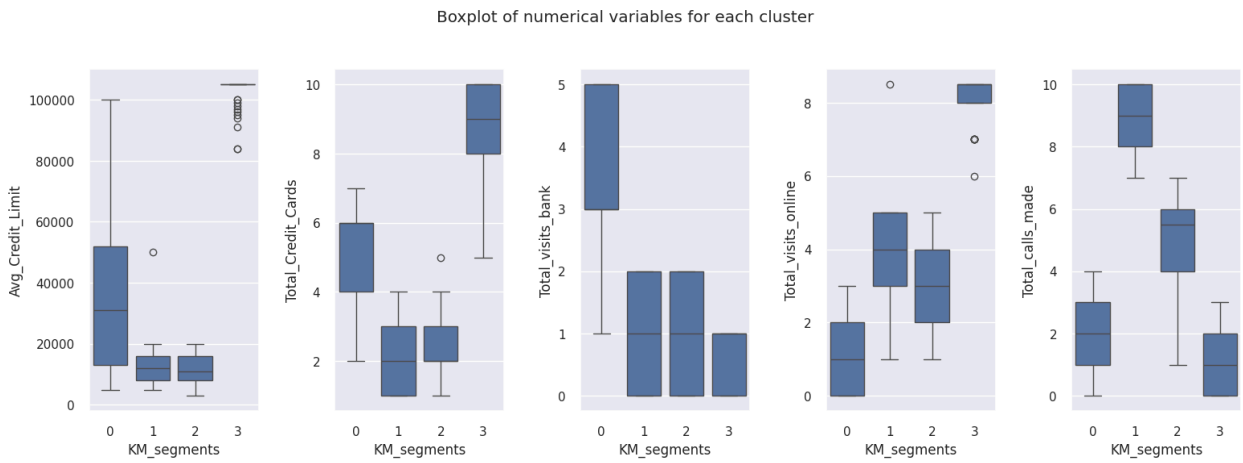


Fig 38: Boxplot of numerical variables for each cluster

Conclusions K-means

- K-Means produced balanced and compact clusters, with customers grouped mainly by credit limit and engagement level.
- Cluster 0 contained high-value, high-engagement customers (frequent bank/online visits, multiple cards).
- Cluster 1 included low-engagement customers with fewer cards and minimal usage.
- Cluster 2 represented average users with moderate credit limits and moderate engagement.
- Cluster 3 grouped very low-engagement or inactive customers, signaling churn risk.

The silhouette score indicated that K-Means formed well-separated clusters with good cohesion.

Hierarchical Clustering

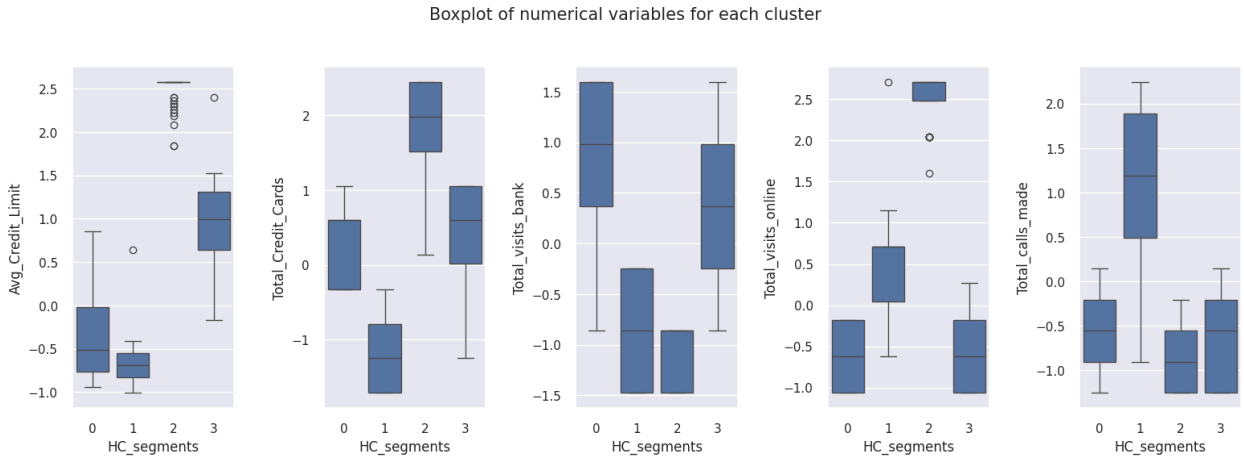
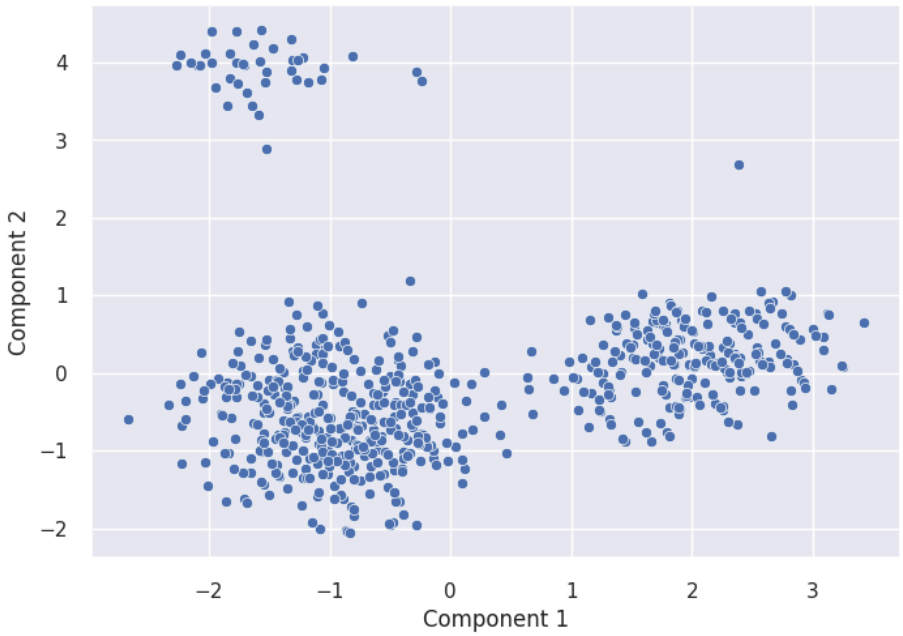


Fig 39: Boxplot Of Numerical Variables For Each Cluster

Conclusions Hierarchical clusters

- Hierarchical clustering also revealed four distinct customer groups with similar behavioral patterns to K-Means.
- Cluster 0 captured premium customers with higher credit limits and high engagement.
- Cluster 1 grouped low-engagement, low-usage customers.
- Cluster 2 included moderate customers with average engagement and potential to grow.
- Cluster 3 identified customers with extremely low or no activity, likely inactive.

Using PCA To Reduce The Number Of Variables



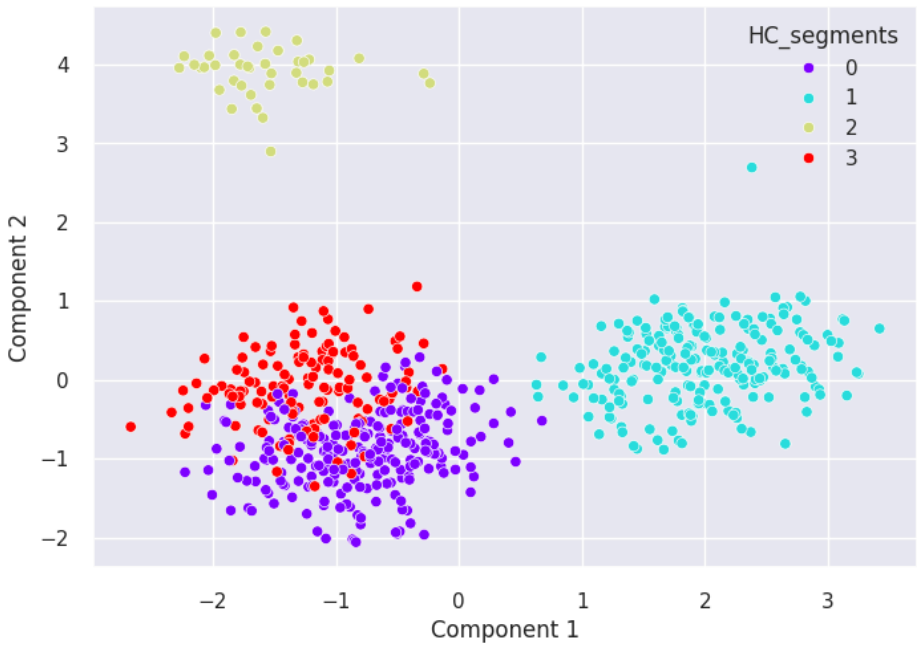


Fig 40: PCA To Reduce The Number Of Variables

The clusters are pretty well separated.

Actional Insights and Recommandations

Actionable Insights

- Customers are naturally segmented into four behavior groups: premium, moderate, low-usage, and inactive.
- Credit limit and number of cards are the strongest differentiators across clusters.
- Bank and online visit frequency reflects customer engagement and separates active vs. inactive users.
- Cluster 0 (Premium/Engaged) shows the highest profitability potential due to high credit use and engagement.
- Cluster 1 (Low Usage) has very few interactions, suggesting low awareness or limited product fit.
- Cluster 2 (Moderate Users) has mid-level engagement, making them easier to convert to premium users.
- Cluster 3 (Inactive/At-Risk) has almost no activity, signaling potential churn.
- K-Means provides balanced, well-separated clusters, suitable for marketing execution.
- Hierarchical clustering highlights customer relationship patterns, though clusters are less balanced.
- Both methods confirm that engagement level directly correlates with customer profitability.

Recommendations

- Retain Cluster 0 through loyalty programs, exclusive rewards, and priority services.
- Activate Cluster 1 using awareness campaigns, educational content, and simplified onboarding.
- Grow Cluster 2 by offering personalized promotions and cross-selling opportunities.
- Reactivate Cluster 3 with win-back campaigns, fee waivers, and re-engagement offers.
- Launch a tiered customer engagement strategy (Retain → Grow → Activate → Reactivate).
- Invest in digital engagement tools (apps, online services) to improve adoption among low-usage clusters.
- Monitor customer movement between clusters to measure effectiveness of interventions.
- Use K-Means clusters for operational marketing because of their balance and clarity.
- Apply Hierarchical analysis for strategic insights, especially to understand customer relationships.
- Regularly retrain the clustering model with new data to keep segmentation current.